

Q1 - 24 June - Shift 1

A projectile is projected with velocity of 25 m/s at an angle θ with the horizontal. After t seconds its inclination with horizontal becomes zero. If R represents horizontal range of the projectile, the value of θ will be : [use $g = 10 \text{ m/s}^2$]

(A) $\frac{1}{2} \sin^{-1} \left(\frac{5t^2}{4R} \right)$ (B) $\frac{1}{2} \sin^{-1} \left(\frac{4R}{5t^2} \right)$

(C) $\tan^{-1} \left(\frac{4t^2}{5R} \right)$ (D) $\cot^{-1} \left(\frac{R}{20t^2} \right)$

Space for your notes:

Q2 - 24 June - Shift 2

A body is projected from the ground at an angle of 45° with the horizontal. Its velocity after 2s is 20 ms^{-1} . The maximum height reached by the body during its motion is _____m. (use $g = 10 \text{ ms}^{-2}$)

Space for your notes:

Q3 - 25 June - Shift 2

Questions

MathonGo

Given below are two statements. One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Two identical balls A and B thrown with same velocity 'u' at two different angles with

horizontal attained the same range R. If A and B reached the maximum height h_1 and h_2

respectively, then $R = 4\sqrt{h_1 h_2}$

Reason R: Product of said heights.

$$h_1 h_2 = \left(\frac{u^2 \sin^2 \theta}{2g} \right) \cdot \left(\frac{u^2 \cos^2 \theta}{2g} \right)$$

Choose the CORRECT answer :

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false

(D) A is false but R is true

Space for your notes:

Q4 - 26 June - Shift 1

A fighter jet is flying horizontally at a certain altitude with a speed of 200 ms^{-1} . When it passes directly overhead an anti-aircraft gun, bullet is

fired from the gun, at an angle θ with the horizontal, to hit the jet. If the bullet speed is 400 m/s , the value of θ will be $^\circ$.

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Questions

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Q5 - 27 June - Shift 1

A projectile is launched at an angle ' α ' with the horizontal with a velocity 20 ms^{-1} . After 10 s, its inclination with horizontal is ' β '. The value of $\tan \beta$ will be : ($g = 10 \text{ ms}^{-2}$)

(A) $\tan \alpha + 5 \sec \alpha$ (B) $\tan \alpha - 5 \sec \alpha$

(C) $2 \tan \alpha - 5 \sec \alpha$ (D) $2 \tan \alpha + 5 \sec \alpha$

Space for your notes:

Q6 - 27 June - Shift 1

A girl standing on road holds her umbrella at 45° with the vertical to keep the rain away. If she starts running without umbrella with a speed of $15\sqrt{2} \text{ kmh}^{-1}$, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is :

(A) 30 kmh^{-1} (B) $\frac{25}{\sqrt{2}} \text{ kmh}^{-1}$

(C) $\frac{30}{\sqrt{2}} \text{ kmh}^{-1}$ (D) 25 kmh^{-1}

Space for your notes:

Q7 - 28 June - Shift 1

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Questions

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Motion of a particle in x-y plane is described by a

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set of following equations $x = 4\sin\left(\frac{\pi}{2} - \omega t\right)$ m and

$y = 4\sin(\omega t)$ m. The path of particle will be –

(A) circular

(B) helical

(C) parabolic

(D) elliptical

Q8 - 28 June - Shift 2

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Questions

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A ball is spun with angular acceleration

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$\alpha = 6t^2 - 2t$ where t is in second and α is in rads^{-2} . At $t = 0$, the ball has angular velocity of 10 rads^{-1} and angular position of 4 rad . The most appropriate expression for the angular position of the ball is:

(A) $\frac{3}{2}t^4 - t^2 + 10t$

(B) $\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$

(C) $\frac{2t^4}{3} - \frac{t^3}{6} + 10t + 12$

(D) $2t^4 - \frac{t^3}{2} + 5t + 4$

Q9 - 29 June - Shift 2

A person can throw a ball upto a maximum range of 100 m . How high above the ground he can throw the same ball?

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(A) 25 m

(B) 50 m

(C) 100 m

(D) 200 m

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Answer Key

Q1 (D)

Q2 (20)

Q3 (A)

Q4 (60)

Q5 (B)

Q6 (C)

Q7 (A)

Q8 (B)

Q9 (B)

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Q1 (10)

$$R = \frac{V^2 (2 \sin \theta \cos \theta)}{g}$$

$$t = \frac{V \sin \theta}{g} \Rightarrow V = \frac{gt}{\sin \theta}$$

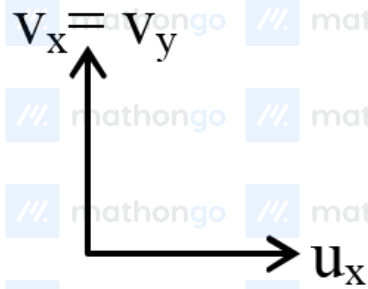
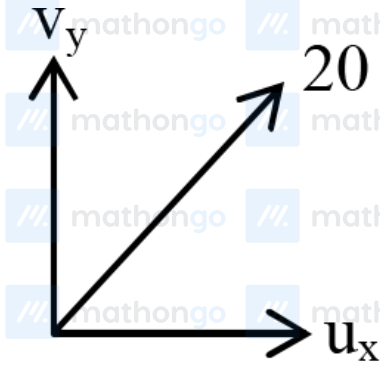
$$\Rightarrow R = \frac{g^2 t^2}{\sin^2 \theta} \cdot \frac{2 \sin \theta \cos \theta}{g}$$

$$\tan \theta = \frac{2gt^2}{R} = \frac{20t^2}{R}$$

$$\cot \theta = \frac{R}{20t^2}$$

Q2 (20)

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$$V_y = V_x - 20$$

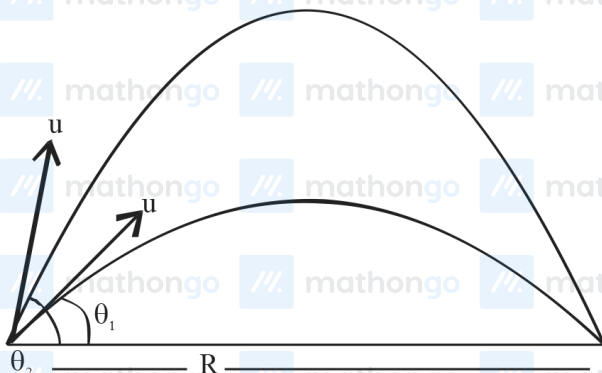
$$\sqrt{(u_x - 20)^2 + u_x^2} = 20$$

$$\Rightarrow 2u_x^2 - 40u_x = 0$$

$$\therefore u_x = 20$$

Q3 (A)

For same range $\theta_1 + \theta_2 = 90^\circ$



$$h_1 = \frac{u^2 \sin^2 \theta_1}{2g} \quad h_2 = \frac{u^2 \sin^2 \theta_2}{2g}$$

$$h_1 h_2 = \frac{u^2 \sin^2 \theta_1}{2g} \times \frac{u^2 \sin^2 \theta_2}{2g}$$

$$\theta_2 = 90 - \theta_1$$

$$h_1 h_2 = \frac{u^2 \sin^2 \theta_1}{2g} \cdot \frac{u^2 \cos^2 \theta_1}{2g}$$

$$= \left[\frac{u^2 \sin \theta_1 \cos \theta_1}{2g} \right]^2$$

$$= \left[\frac{u^2 \sin \theta_1 \cos \theta_1}{2g} \times \frac{2}{2} \right]^2 = \frac{R^2}{16}$$

$$R = 4\sqrt{h_1 h_2}$$

So R is correct explanation of A

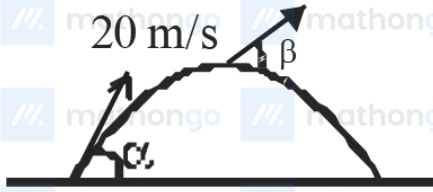
Q4 (60)

Both should have same horizontal component of velocity

$$200 = 400 \cos \theta$$

$$\theta = 60^\circ$$

Q5 (B)



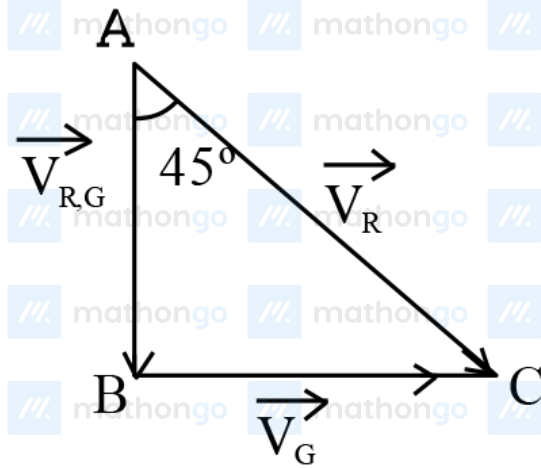
$$v_x = u_x = 20 \cos \alpha$$

$$v_y = 20 \sin \alpha - 10 \times 10$$

$$\tan \beta = \frac{v_y}{v_x} = \frac{20 \sin \alpha - 100}{20 \cos \alpha}$$

$$= \tan \alpha - 5 \sec \alpha$$

Q6 (C)



$$V = \tan \theta = \frac{V_G}{V_{RG}}$$

$$1 = \frac{V_G}{V_{RG}} \Rightarrow 15\sqrt{2} = V_{RG}$$

Q7 (A)

$$x = 4 \sin\left(\frac{\pi}{2} - \omega t\right) \quad y = 4 \cos(\omega t)$$

$$x = 4 \cos(\omega t) \quad y = 4 \sin(\omega t)$$

Eliminate 't' to find relation between x and y

$$x^2 + y^2 = y^2 \cos^2 \omega t + y^2 \sin^2 \omega t = 4^2$$

$$x^2 + y^2 = 4^2$$

Q8 (B)

$$\frac{dw}{dt} = 6t^2 - 2t$$

$$\int_{10}^w dw = 2t^3 - t^2$$

$$w = 10 + 2t^3 - t^2$$

$$\frac{d\theta}{dt} = 10 + 2t^3 - t^2$$

$$\int_4^\theta d\theta = 10t + \frac{t^4}{2} - \frac{t^3}{3}$$

$$\theta = 4 + 10t + \frac{t^4}{2} - \frac{t^3}{3}$$

Q9 (B)

$$R = \frac{u^2 \sin 2\theta}{g} \quad R_{\max} = \frac{u^2}{g} = 100$$

$$H_{\max} = \frac{u^2}{2g} = \frac{100}{2} = 50\text{m}$$

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